

Information and communications technologies to facilitate concurrent engineering in construction

C J Anumba, G Baron and A Duke

The adoption of concurrent engineering techniques in the construction industry needs to be underpinned by cost-effective, secure and reliable communications networks. These need to support a wide range of communications facets which are considered important in concurrent life cycle design and construction (CLDC). This paper investigates the enabling information and communications technologies which are applicable in this context. It briefly describes these technologies highlighting, where appropriate, current developments and future trends. The extent to which the identified facets of communication are addressed by the new and emerging technologies is also examined and recommendations made on the way forward.

1. Introduction

Construction projects normally involve transient ‘virtual organisations’ made up of members of the project team working together on the design and construction of a facility. A typical project involves several disciplines; for example, the participants in a building project may include the client, the architect, the structural engineer, the building services engineer, the quantity surveyor, the main contractor, sub-contractors, materials suppliers, and other specialists. Many of these participants often work independently while taking decisions that inevitably affect others. The adoption of concurrent engineering principles by the construction industry is increasingly being seen as vital for reducing the problems posed by the fragmentation of the industry, and enhancing its competitiveness [1]. This involves the integration of all project team members, as well as all the stages in the project life cycle, with a view to ensuring that all key life cycle issues are addressed early in the design process [2]. The benefits of this approach include reduced costs, reduced project durations, greater integration of design information, improved quality, and greater client satisfaction, among others.

An important aspect of concurrent engineering in construction is the need for effective communication of design information at all stages of the constructed facility’s life cycle. Good communications are recognised as vital for virtual organisations (such as construction project teams) to achieve their goals [3]. The key communications facets which need to be addressed in concurrent life cycle design and construction have been discussed elsewhere [4] and are illustrated in Fig 1. They include the following:

- communication between intra-disciplinary CAE tools (Level 1),
- communication between all project team members and their design tools (Level 2),
- communication between design team members (Level 3),
- communication between each discipline and the common project model (Level 4),
- communication across the stages in the project life-cycle (Level 5),
- communication between the project team and third parties (Level 6),
- communication between inter-disciplinary CAE tools (Level 7).

It is clear from the nature of the communications facets highlighted above that paper-based communication of design information is now inadequate to cope with the high level of functionality (in terms of speed, accuracy, usability, ease of modification, enhanced visualisation, consistency, improved co-ordination, etc) required in a collaborative design environment. This is even more so within the context of rapid developments in information and communications technologies, as well as the global environment in which many construction projects are now undertaken. This paper reviews the existing and emerging information and communications technologies which can facilitate concurrent

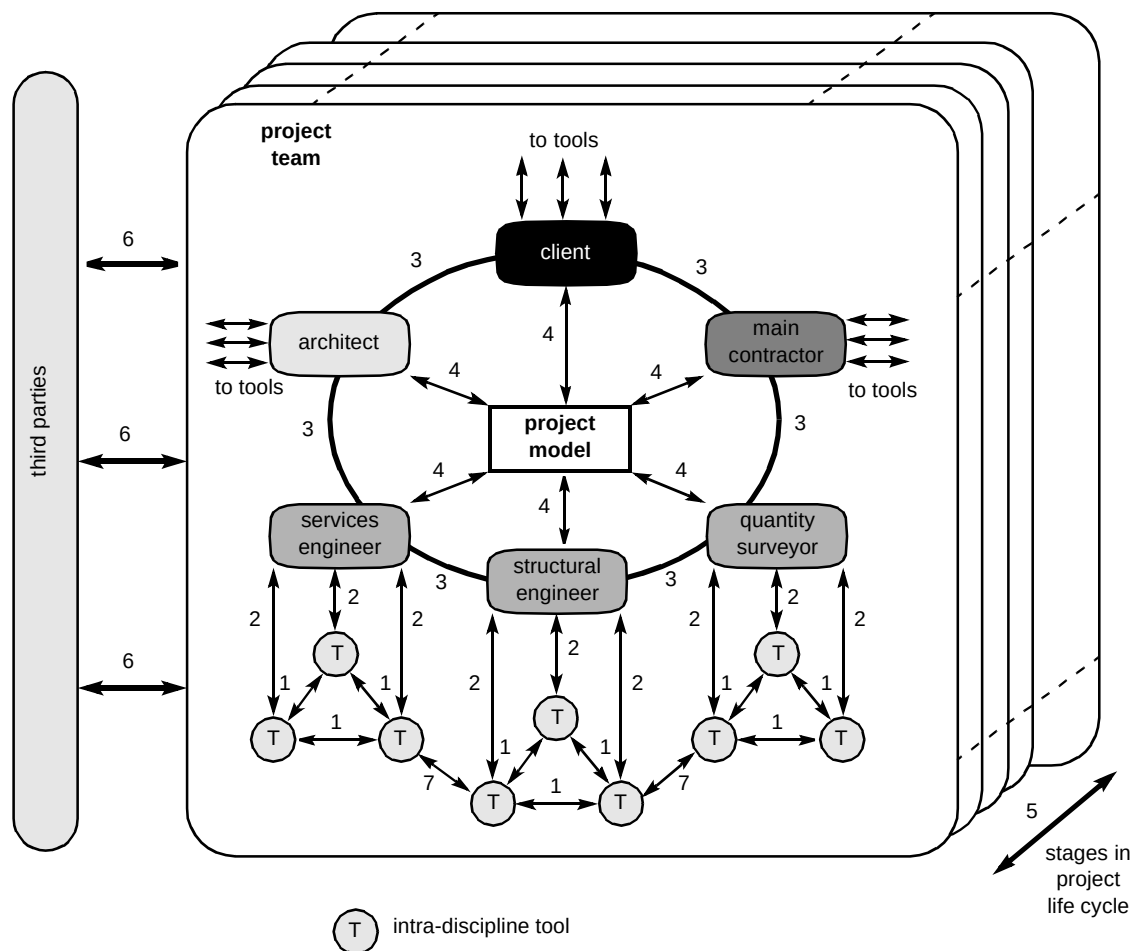


Fig 1 Communication facets in CLDC.

engineering in construction through addressing the identified communications facets.

2. Enabling information and communications technologies

There are several existing and emerging information and communications technologies which have the potential to transform communications within the construction industry. These represent some of the newer information and communications technologies, and are additional to well-established communications tools such as the postal system, telephone, facsimile machines, telex, etc. Rogers [5] sees the potential for these advanced technologies to create collaborative environments that not only enable members of a project team to work together as naturally as possible, but are better than face-to-face meetings. These technologies are briefly described below prior to a discussion of how well they address the identified communications facets.

2.1 Electronic mail

Electronic mail (e-mail) is now widespread and provides a cheap, fast and effective means of communication between two or more people linked together on a computer network. Transmission of messages usually

takes place in seconds or minutes, and there is the facility to 'attach' files (in any mutually acceptable format) to the basic mail message. Information transmission is reasonably secure and most electronic mailing systems provide facilities for delivery confirmation, auto-reply, mail filtering, and priority setting.

2.2 Desktop videoconferencing

Sometimes referred to as videotelephony, this is now fairly well established and is already being used to facilitate collaboration on many complex projects. It involves the provision of a video and audio link between two or more parties who may be co-located or geographically distributed. It also provides facilities for 'whiteboard sharing' allowing groups to transfer, co-view, discuss and co-edit data files (containing textual and/or graphical information). Many systems also support application sharing, file transfer, and textual communication which can be saved as a record of the conference.

2.3 Multimedia

Multimedia communications systems integrate sound, pictures (still and/or moving), and data (textual and/or graphical). The aim is to broaden the communications bandwidth between the end-user and the system, thereby

enhancing the quality of the user interface. Many of the other communications technologies described in this paper (e.g. videoconferencing) take advantage of multimedia technology.

2.4 The Internet

The Internet is an international network of computers which are geographically distributed but are able to exchange and communicate information. The best known form is the World Wide Web, WWW (also known as 'the Web') which is certainly the fastest growing repository of information available. Organisations and individuals have the facility to create their own websites (also known as 'home pages') containing whatever information they wish to place in the public domain. Using appropriate 'plug-ins', it is also possible to import and view 2-D or 3-D CAD models within the 'browser'. This has significant potential for information delivery between project team members across wide geographical locations. Future Web 'browser' developments planned by Netscape and Microsoft are expected to further enhance collaborative working via the Internet.

2.5 Intranets

It is sometimes necessary for groups of people to communicate and share information in such a way that only members of the group have access to the information. This is achieved through the use of 'intranets', which are business information networks that are 'firewall' protected and link members of the group together. The group could consist of people within an organisation or members of a 'virtual organisation' set up to collaborate on a project. It may be necessary to provide access to the Internet (or other networks) from an intranet.

2.6 Virtual reality

Virtual reality (VR) systems provide users with a three-dimensional view of a computer-generated world [5]. VR offers the potential for highly realistic visualisations of 3-D spaces and objects in a way not possible with 2-D representations or 3-D CAD models. Both immersive and non-immersive VR models are available. With non-immersive VR, the user is able to navigate the 3-D VR model using conventional keyboard and mouse input devices. Immersive VR, on the other hand, seeks to place the user within the same space as the data being visualised and/or manipulated. It requires more sophisticated devices including special head-mounted viewing equipment and sensory data-gloves (Fig 2).



Fig 2 Use of head-mounted display and data glove.

Virtual worlds can be created on either PC or workstation platforms using a combination of VRML (virtual reality modelling language), proprietary developer's toolkits, and CAD models. The use of a portable 3-D digital theatre called the 'VisionDome' enhances collaboration by enabling ten or more people to be concurrently immersed in a VR environment without the constraints of head-mounted displays, goggles or gloves.

2.7 Mixed reality

Mixed reality (MR) environments seek to combine the best features of real environments with those of virtual environments. There are perceived benefits in this mixture, with the two main types of mixed reality environments being augmented reality (AR) and augmented virtuality (AV), depending on whether the environment that is being augmented is primarily 'real' or 'virtual'. Figure 3 illustrates the relative position of these environments in what Milgram and Kishino [6] term 'The Virtuality Continuum'.

2.8 Virtual meeting room

Virtual meeting rooms (VMR) represent an extension of the concept of desktop videoconferencing. In a virtual meeting room, team members are able to interact intuitively

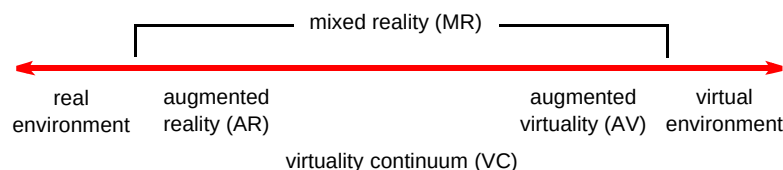


Fig 3 The virtuality continuum [6].

in 3-D space and feel as though they were all in the same room. This is considered to be more realistic than desktop conferencing but requires the use of appropriate metaphors and/or avatars to represent real-world objects and people. VMR technology is still in its infancy and many systems do not at present support the full range of required interactions or realistic pictorial representation of the parties present in a meeting (Fig 4).

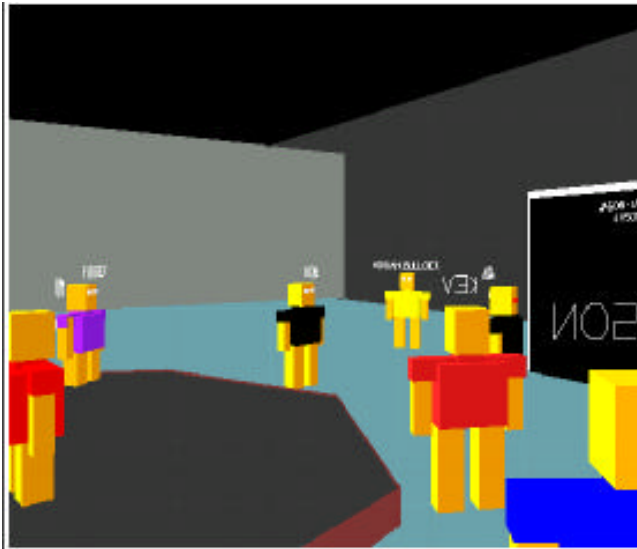


Fig 4 Virtual meeting room (VMR).

2.9 Shared 3-D project model

As shown in Fig 1, a 3-D shared project model is central to concurrent project development, and communication between team members. Within a shared environment, members of a team can concurrently view a 3-D model, discussing and agreeing changes. The common 3-D project model has the potential for use at all stages in the life cycle of a construction project and, as the single complete repository of project information, will ensure that consistency is maintained. This is discussed further later in the paper.

2.10 People and Information Finder (PIF)

The People and Information Finder (PIF) incorporates the AP-IA (Availability of Persons — Intelligent Agent) system developed in the Virtuosi project [7, 8]. It is designed to aid communications between people at remote sites by estimating the likelihood of a person's availability. As shown in Fig 5, the principal sources of the availability information are e-mail usage, telephone usage, electronic diary entry, computer activity and a video still (based on a miniature video camera mounted to provide an overview of a person's work area). PIF evaluates the information from the various sources and summarises the person's availability.

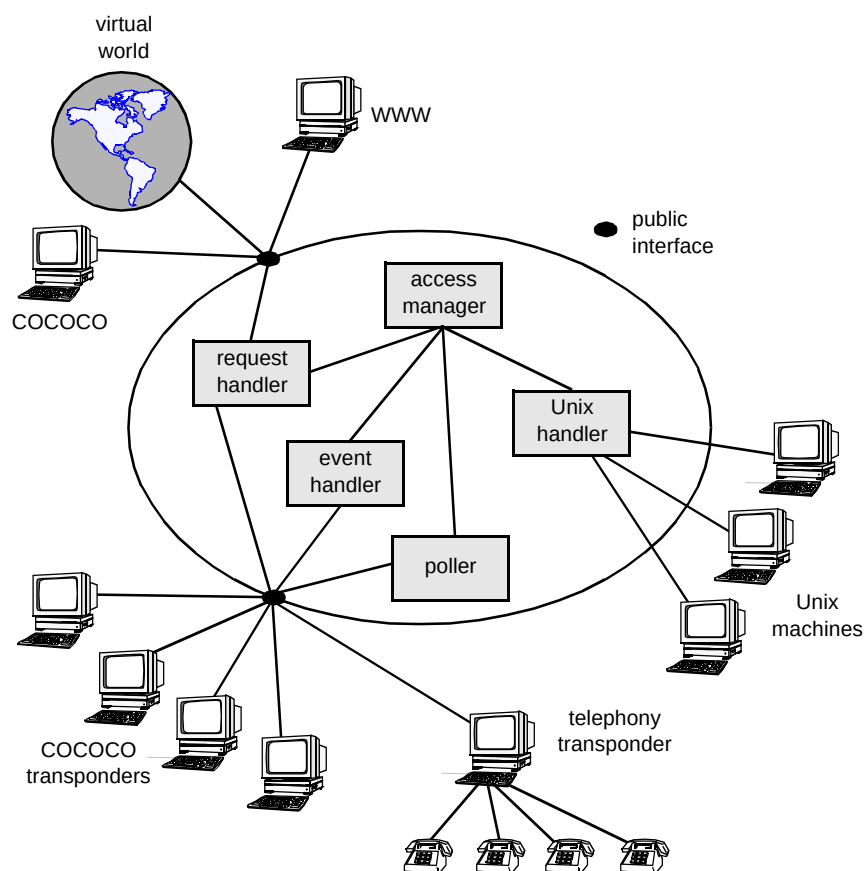


Fig 5 PIF architecture [7].

2.11 Broadband communications networks

Broadband communications networks combine digital transmission and switching capabilities to facilitate mixing together different types of traffic — voice, video and data [9]. One such network is the broadband integrated services digital network (B-ISDN) which supports telephony, video-telephony, database access, file transfer, shared-screen working, electronic meetings and high-quality video transmission. B-ISDN allows for dynamic bandwidth allocation in accordance with user requirements; this ensures optimum utilisation of the available bandwidth to provide high quality images and fast transmission times. Transmission is based on the asynchronous transfer mode (ATM) in which all information to be transmitted is first fragmented into small, fixed-sized frames known as cells [10].

2.12 Data exchange protocols

Electronic data interchange has long been a problem in collaborative working. With the fragmented nature of the construction industry and the number of disciplines represented on construction project teams, the need for a solution to the problem is even more acute. Significant progress has been made with the use of DXF (from Autodesk) which has emerged as the *de facto* data exchange protocol in the construction industry. Further developments in electronic data exchange are expected to be based on emerging standards such as STEP (Standard for the Exchange of Product Data) and the Industry Foundation Classes (IFCs) — common class definitions for objects — being developed under the auspices of the International Alliance for Interoperability (IAI).

2.13 Electronic document management systems

Electronic document management (EDM) systems are sets of software and services through which business-critical information is managed by enabling the creation, assembly, control, reuse, and distribution of this information [11]. The information managed may be in different electronic formats — text, graphics, images, CAD models, video, audio, etc. EDMs help to improve communications both within and between organisations by improving accuracy, retrieval and flow of information. A recent EDM system which has potential for use in the construction industry is BAIRS (BT Automated Information Retrieval System). Based on Lotus Notes, BAIRS was initially developed as a drawing management system but is being extended to cover all documents. In addition to other features, it provides for controlled access, change and comment management, and an audit trail.

2.14 Mobile communications systems

These seek to facilitate the communication of multimedia information between geographically distributed locations. One of the notable developments in this regard is the multimedia hard hat, MMHH (Fig 6) which was intended to improve communications between the construction site and the design office [12]. A site worker wearing the MMHH is able to view plans and information held electronically on the hat-mounted display, while design office personnel can monitor developments on site through video images sent back by the video camera mounted on the hard hat.



Fig 6 MultiMedia Hard Hat.
[© 1995-1996 BICC plc]

3. Mapping of technologies to communications facets

The above information and communications technologies obviously have a role in the development of a communications infrastructure to facilitate concurrent engineering in construction. The specific potential application of each of the identified technologies is examined below in relation to the different facets of communication outlined earlier. In many cases two or more technologies are applicable.

3.1 Intra- and inter-disciplinary tool/tool communication

These levels of communication are more a function of the individual tools and their associated hardware platforms, and so are less readily supported by many of the information and communications technologies discussed above. Data exchange protocols (such as DXF, STEP [13], etc) offer the greatest potential for addressing tool/tool

communication and will enable design information to be transmitted between different applications and between different disciplines. It is important that the exchange protocol adopted is able to convey both the underlying context and semantics of the design information. In the case of tool/tool communication in a distributed environment, appropriate communications networks based on LANs (wireline or wireless), WANs and B-ISDNs also have a role.

3.2 Designer/tool communication

The quality of communication between designers and their tools is highly dependent on the design of the user interfaces to the tools. Clearly, the more designer-oriented a system is, the more efficiently it can be deployed by the designer. The advent of multimedia, virtual reality and mixed reality offer much scope for increasing the communication bandwidth between designers and their tools, thereby improving the quality of user/system interaction. Many computer-aided engineering (CAE) systems already employ multimedia techniques and offer enhanced visualisation of design information. These are expected to lead to a better understanding of the operation of these systems and hence, more effectiveness, improved throughput and better quality [14].

3.3 Project team communication

This level of communication is the one on which the emerging information and communications technologies are likely to have the most impact. Given the fragmentation problem in the construction industry, there is potential for significant cultural changes. All the listed technologies do have a role in this regard albeit to varying extents as discussed below.

- Electronic mail and videoconferencing, although well-established in the global context, are still used only to a limited extent by construction industry practitioners. E-mail can be used to convey design information (textual and graphical) across vast distances in a matter of seconds while videoconferencing will enable project team members to have videotelephony links which will enable discussion of design and/or construction issues.
- Multimedia systems will improve the richness of communication between project team members, and will increasingly be embedded in the other communications systems, as well as in CAE tools.
- The Internet represents not only an invaluable source of information for construction project teams, but an inexpensive means of communication between design team members. It also offers an economical advertisement platform for clients, consultants,

contractors and material suppliers, and provides the opportunity to set up websites to 'showcase' both ongoing and past projects.

- An intranet offers the construction project team secure communications links between all the appropriate parties on a project. However, it may be necessary to temporarily allow limited access to third-party specialists.
- Both virtual and mixed reality can be used to improve communications and co-ordination between members of a construction project team. The ability to totally immerse the team within a VR environment will enhance visualisation of the project, facilitate interaction, and provide a better 'feel' for the working of a facility. Being able to superimpose CAD or VR models of buildings on real scenes or vice versa has obvious visualisation benefits for the project team, and the construction client in particular.
- A virtual meeting room will improve communications between project team members and reduce travel to project team meetings. However, it will have a greater impact where links are provided to VR and/or MR models of the project or construction site.
- The potential role of a shared 3-D project model in improving project team communications is well recognised, and has immense benefits in terms of model consistency and improved co-ordination. Appropriate mechanisms for access, change and version management will need to be established.
- PIF also has tremendous potential for improving communication. It will facilitate location of, and communication with, appropriate members of the project team at all stages of the project life cycle. Privacy and security issues in its use do need to be addressed beforehand.
- B-ISDN is vital for facilitating the fast transmission of rich multimedia information between members of a project team across both short and vast distances.
- Data exchange protocols can enhance project team communications by facilitating the direct exchange of design information between their heterogeneous CAE systems.
- Electronic document management (EDM) systems are well-suited to managing the flow of information between the disparate professionals in construction project teams. The large volume of documents generated by construction projects can be effectively and securely managed using an appropriate EDM system.

- Mobile communications systems are very important for bi-directional communication between (remote) sites and offices. They are likely to be particularly useful for communication between the contractor on site and design consultants (or the client). Site investigation teams at the early stages of construction also stand to benefit significantly from these systems.

3.4 Discipline/project model communication

Bi-directional communication between each discipline and the central project model is very important. Most of the information and communications technologies discussed have a role in this regard. E-mail, the Internet, an intranet, and B-ISDN will facilitate access to the project model by each member of the project team. The format of design information, and exchange or visualisation mechanisms, will depend to a large extent on the hardware and software platforms of both the project model and the intra-disciplinary CAE tools. Discipline partial-models may be of CAD, VR or MR format but it is important that compatibility is ensured. This may be achieved either by all members of the project team using the same platforms (which is restrictive) or using systems that support an appropriate, effective and mutually acceptable data exchange format.

3.5 Communication between stages in the project life cycle

This facet of communication is one of the most important but is only marginally supported by existing CAE systems and communications protocols. Perhaps, the most important technology in supporting communication between stages in the project life cycle is the common 3-D project model. The shared model can store both graphical and non-graphical design information which can be updated as the project progresses, allowing information from the early stages of the project to be available for reuse at later stages. For maximum benefit, the non-graphical information stored in the model should include details of design intent and rationale in such a way that these are traceable to explicit and implicit client requirements. In addition, safety information (possibly, encapsulated in the Health and Safety Plan and the Health and Safety File required under the CDM regulations [15]) should be included in the project model. It is also essential that adequate consideration is given to archiving methods to ensure that the project model does not become obsolete within a short time-scale.

3.6 Project team/third party communication

All the information and communications technologies identified earlier as beneficial for communication between project team members are equally applicable to this facet of communication. The only difference lies in the fact that

third parties will only need to have limited and short-term access to the project model and the project team's communications network. Once their input to the project is complete, such third parties will lose their access rights. It is important that details of these third parties are stored within the people information database (PID) so that, where necessary, some of them may be reconnected at a later date if their services are required.

4. Discussion

It is clear from the foregoing that there is considerable potential for the use of emerging information and communications technologies to support concurrent engineering within the construction industry. While some facets of communication can be supported by several of these technologies, others are less well supported. For example, there is much scope for transforming communications between members of a project team, and between a project team and third parties. This should bring project team members closer together and help to reduce the fragmentation within the construction industry. Other benefits of the information and communications technologies discussed include:

- easier access to appropriate project team personnel, third parties and information,
- fast and efficient access to design and other project information,
- secure communication of design and construction information within a project team,
- enhanced visualisation of design information and the project as a whole,
- more realistic communication of design intent and rationale to clients and other members of the project team,
- improved co-ordination of inter-disciplinary perspectives,
- increased consistency of project information,
- reuse of project information from other disciplines or project stages without the need for re-input,
- improved corporate knowledge due to centralised pool of project information and improved communications protocols.

Aspects of project team communication which are not adequately supported include those which are primarily software and/or platform-dependent (i.e. levels 1, 2, 4 and 7). Intra- and inter-disciplinary tool/tool communication (levels 1 and 7) and discipline/project model communication (level 4), as mentioned earlier, need to be based on the use of common software and hardware platforms or appropriate data exchange protocols which allow smooth

information interchange between heterogeneous systems. Designer/tool communication (level 2) can be significantly improved by better user-interface design and the use of enhanced visualisation and simulation tools such as VR, MR and multimedia.

The use of a shared 3-D project model is central to the success of concurrent engineering in construction. It should be noted that no existing system possesses the level of functionality required to adequately support the 'seamless inter-working' possible with such a model. In particular, there are limited facilities for ensuring the persistence of design intent and rationale. Also, the provisions in many systems for change propagation and management, and version control are very limited. However, there are real prospects for the emergence, in the short to medium term, of modelling systems with the required functionality.

In addition to adopting these emerging information and communications technologies, there is a need to ensure that these are effectively managed to ensure maximum benefit. All project team members need to agree guidelines for the use of the communications infrastructure when set up, and clear mechanisms established for access, change and version management.

5. Conclusions

Effective communication is necessary for ensuring the success of concurrent engineering in construction. This paper has examined key emerging information and communications technologies which have a role to play in delivering this. These include electronic mail, desktop videoconferencing, multimedia, the Internet, intranets, virtual reality (VR), mixed reality (MR), virtual meeting rooms, shared 3-D project model or workspace, People and Information Finder (PIF), broadband communications networks, data exchange protocols, electronic document management systems, and mobile communications systems. The extent to which these technologies address the communication facets identified earlier [4] is also discussed in detail. The most obvious mode of communication to which most of these can be applied is person-to-person communication — intra-project team or project team/third party communication. There is also some support for person-to-computer and computer-to-computer communication.

The greatest potential for radically transforming communications in concurrent life cycle design and construction lies in the use of a common 3-D project model with capabilities for 'seamless' multi-user working. Among other things, the common project model should support:

- individual discipline interactions with the central model,
- heterogeneous intra-discipline tools,
- configuration management,
- perpetuation of design intent and rationale across stages in the project life cycle,
- emerging standards for information representation, interchange and interoperability,
- integration with a robust and multi-faceted project communications infrastructure,
- enhanced visualisation of design and construction processes based on multimedia, virtual and mixed reality, simulations, video, etc,
- an open architecture to facilitate extensions and customisation to suit individual project and team requirements.

Clearly, there are many communications issues inherent in adopting concurrent engineering principles in construction. These all need to be effectively addressed if the full benefits of concurrent project development are to be realised. It can be seen from this paper that the enabling information and communications technologies are either in existence or emerging. These have the potential to enable distributed construction project teams to work as effectively, if not more so, than co-located teams engaged in face-to-face meetings. The European project, Collaborative Integrated Communications for Construction (CICC), is playing a major part in the development and evaluation of many of these technologies within a construction context.

Acknowledgement

BT has sponsored the project on which this paper is based under its short-term research fellowship scheme.

References

- 1 Anumba C J, Evbuomwan N F O and Sarkodie-Gyan T: 'An approach to modelling construction as a competitive manufacturing process', in DeAlmeida S M (Ed): 'Competitive Manufacturing', Proceedings, 12th Annual Conference of the Irish Manufacturing Committee, Cork, Ireland, pp 1069—1076 (September 1995).
- 2 Evbuomwan N F O and Anumba C J: 'Towards a concurrent engineering model for design and build projects', *The Structural Engineer*, **74**, No 5, pp 73—78 (1996).
- 3 Rogers A S: 'Virtual reality — the new media?', *BT Technol J*, **13**, No 4, pp 132—137 (October 1995).
- 4 Anumba C J, Baron G and Evbuomwan N F O: 'Communications issues in concurrent life cycle design and construction', *BT Technol J*, **15**, No 1, pp 209—216 (1997).
- 5 Rogers A S: 'Virtuosi — virtual reality support for groupworking', *BT Technol J*, **12**, No 3, pp 81—89 (July 1994).
- 6 Milgram P and Kishino F: 'A taxonomy of mixed reality visual displays', *IEICE Transactions on Information Systems*, **E77-D**, No 12 (December 1994).
- 7 Duke A: 'Finger server issues', Internal BT Report (April 1996).

- 8 Tracey I: 'Structure, networking and presentation issues', Internal BT Report (April 1996).
- 9 Popple G.: 'Networks for BRICC', Construction Industry Computer Exhibition (CICE) (February 1993).
- 10 Halsall F: 'Data communications, computer networks and open systems', Addison-Wesley, Wokingham, UK, 4th Ed (1996).
- 11 Interleaf: 'The document management guide', published on the Internet — URL: <http://www.ileaf.com/docman.html> (1996).
- 12 Thorpe A et al: 'Multimedia communications in construction', Civil Engineering, Proc Instn of Civ Engrs, 108, pp 12—16 (February 1995).
- 13 Crowley A and Watson A: 'A guide to painless computer integration', The Structural Engineer, 74, No 3, pp 50—52 (1996).
- 14 Anumba C J and Watson A S: 'An innovative approach towards designer-oriented CAD systems', The Structural Engineer, 70, No 9, pp 165—169 (1992).



Chimay Anumba graduated in 1984 with a first class honours degree in Building from the University of Jos, Nigeria. He worked briefly as a site engineer and a design engineer before undertaking research in computer-aided engineering at the University of Leeds. On completion of his PhD in 1989, he joined Curtins Consulting Engineers plc and was involved in a wide range of civil and structural engineering projects. He is currently a Senior Lecturer in Civil Engineering and Building at the University of Teesside. Between July and September 1996, he was a short-term research fellow at

BT Laboratories.

- 15 HSE: 'The Construction, Design and Management (CDM) Regulations', Health & Safety Executive (1995).



Godwin Baron joined BT as a technician in 1987. Following duties as a network analyst in Colchester Computer Network Management Centre in 1992, he became Information Systems Manager for the Corporate Research Programme, developing and deploying systems to support the Corporate Research Programme.

For the last year he has been involved with the European Collaborative Project CICC.



Alistair Duke graduated from Aston University in 1994 with an MEng in Electronic Systems Engineering. Since joining BT he has worked on the collaborative projects, Virtuosi and CICC (Collaborative Integrated Communications for Construction). Both of these have been concerned with using virtual reality and collaborative communications to support groupworking.

He is a member of the Centre for Human Communication at BT Laboratories and manager of the CICC project.